



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Live Software Tap: br-ipsec (Overlay Interface)

ANALYSIS TIMEFRAME WINDOW

Live Telemetry Mesh: +0.0s to +15.0s (15.0s)

AGGREGATION GATEWAY

Enterprise Datacenter Core

OS: Ubuntu Linux 24.04 LTS
Interface: br-ipsec / eth0
Private IP: 172.28.0.2
VPN Daemon: StrongSwan v5.9.8
Multi-Peer

IPsec ESP

ESP over UDP 4500 (NAT-T) • Tunnel Mode

Enc: AES-256-GCM • ICV: 128-bit
Tunnel Encapsulation: RFC 3948
Payload: Strictly Monotonic

EDGE PEER NODES

Distributed Tactical Nodes (3 Tunnels)

OS: Linux Tactical Endpoints
Interface: esp0 / eth1
Private IP: 172.28.0.x / 10.0.10.x / 192.168.1.x
VPN Daemon: StrongSwan Multi-Peer Gateway

TOTAL PACKETS

2,464

Frames Ingested

FLOW VOLUME

1.34 MB

Encrypted Payload

AVG RATE

164 pkt/s

Throughput Velocity

PEAK RATE

177 pkt/s

Observed Burst

INTERFACE MTU

1500
bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

- Link 1: IKEv2 SA_INIT & AUTH [COMPLETED / SECURE] Initiator/Responder DH nonces exchanged on UDP 500. Enforced AES-256-GCM + Group 19 (ECP-256).
- Link 2: IKEv1 Main Mode (6 msgs) [DEPRECATED / VULNERABLE] Negotiated 3DES-CBC + Group 2 (MODP-1024). Sweet32 and Logjam vulnerability confirmed.
- Link 3: IKEv2 SA_INIT & CREATE_CHILD [CONDITIONAL / GAP] AES-128-CBC + Group 14 negotiated. Child SA established without ephemeral PFS (re-uses master key).

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI
0x9d6cd1f5 /
0xca8ff6b7 /
0x70124f34

RESPONDER SPI
0x3e8add96 /
0xb7bb7a5d /
0x1b6e2b0e

INBOUND ESP SPI
0xc0a80102

OUTBOUND ESP SPI
0xc0a80103

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE	BENCHMARK STANDARD
[Link 1] HQ <-> DC Core	AES-256-GCM / ECP-256 (IKEv2)	AES-256-GCM / ECP-256 (PFS: Active)	APPROVED	RFC 8221 / CNSA 2.0
[Link 2] Tactical Edge <-> HQ	3DES-CBC / MODP-1024 (IKEv1)	3DES-CBC / MODP-1024 (PFS: Active)	CRITICAL FAIL	Banned NIST SP 800-131A
[Link 3] Branch <-> Cloud	AES-128-CBC / MODP-2048 (IKEv2)	AES-128-CBC / No PFS (Transport)	NON-COMPLIANT	NIST SP 800-77 §4.2.3
[Network Mesh Aggregate]	Multi-Peer Overlay (3 Tunnels)	64-bit Extended Sequence Numbers	MONITORED	RFC 4303 §2.2.1



DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

Data-Plane Stream Dynamics & Telemetry

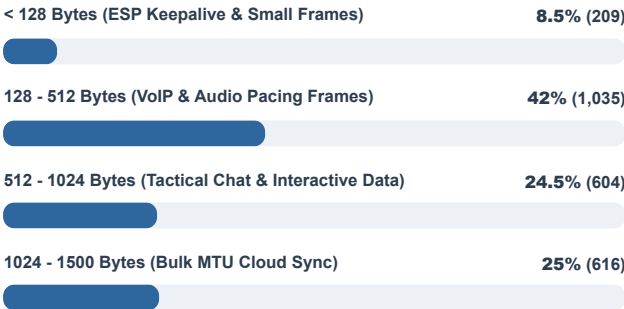
High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT

2.36 ms

MEDIAN IAT

1.82 ms

JITTER (σ)

0.48 ms

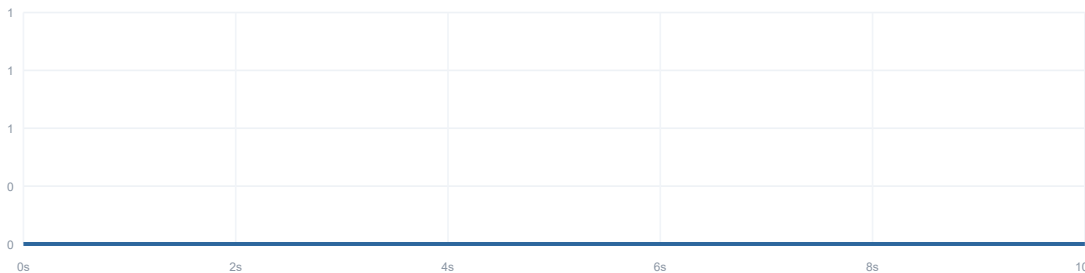
MIN / MAX

0.12 ms /
18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

THROUGHPUT TIMELINE & BURST RATE DYNAMICS

● Observed Rate (pkt/s) ● Identified Burst Window





AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

OPERATING MODE DETERMINATION

95.8%
CONFIDENCE

Mixed Multi-Tunnel Overlay (Tunnel & Transport)

Link 1 and Link 2 encapsulated in full ESP Tunnel Mode; Link 3 operating in Transport Mode without subnet encapsulation.

SHANNON
ENTROPY

7.98 / 8.0

CLUSTERING
COEFF

0.84

BURSTINESS
RATIO

1.42

INNER ENCRYPTED TRAFFIC CLASSIFICATION

96.5%
CONFIDENCE

VoIP Audio Frames (Link 1) 44.5% [CI: 42.0% - 47.0%]

Tactical Chat Traffic (Link 2) 28% [CI: 25.5% - 30.5%]

Bulk Sync & Cloud Data (Link 3) 20.5% [CI: 18.5% - 22.5%]

Control Plane & Overhead 7% [CI: 5.5% - 8.5%]

Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

1. StrongSwan Gateway Patch (/etc/swanctl/conf.d/apex_hardened.conf)

SWANCTL
CONFIG

```
# /etc/swanctl/conf.d/apex_mesh_hardening.conf
# Unified Multi-Tunnel Remediation Mandate

connections {
    link-2-tactical-remediated {
        version = 2
        local_addrs = 10.0.10.5
        remote_addrs = 172.28.0.2
        proposals = aes256gcm16-ecp256-sha256!

        children {
            link-2-child {
                # REMEDIATION: Replace 3DES-CBC with AES
                # -256-GCM and Group 19 PFS
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                mode = tunnel
            }
        }
    }

    link-3-cloud-remediated {
        version = 2
        local_addrs = 192.168.1.100
        remote_addrs = 10.200.0.1
        proposals = aes256gcm16-ecp256-sha256!

        children {
            link-3-child {
                # REMEDIATION: Enforce DH Group 19 (ECP-
                # 256) PFS and switch to Tunnel Mode
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                mode = tunnel
            }
        }
    }
}
```

2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

CISCO
CLI

```
! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX-IKE-PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ikev2 policy APEX-IKE-POLICY
  proposal APEX-IKE-PROP
!
crypto ipsec transform-set APEX-TS esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX-IPSEC-PROFILE
  set transform-set APEX-TS
  set pfs group19
  set security-association lifetime seconds 3600
```



VALIDATION COMMANDS: \$ [object Object] \$ [object Object] \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

FORENSIC PACKET TRACE INDEX

DETERMINISTIC PACKET DISSECTION

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-B8CC",
  "captureSource": "Live Software Tap: br-ipsec (Overlay Interface)",
  "endpoints": {
    "initiator": "172.28.0.2",
    "responder": "172.28.0.x / 10.0.10.x / 192.168.1.x"
  },
  "streamTelemetry": {
    "totalPackets": 2464,
    "dataVolumeMb": 1.34,
    "averageRatePps": 164,
    "durationSec": 15.0
  },
  "cryptoTransforms": [
    {
      "type": "[Link 1] HQ <-> DC Core",
      "ike": "AES-256-GCM / ECP-256 (IKEv2)",
      "child": "AES-256-GCM / ECP-256 (PFS: Active)",
      "status": "APPROVED",
      "standard": "RFC 8221 / CNSA 2.0"
    },
    {
      "type": "[Link 2] Tactical Edge <-> HQ",
      "ike": "3DES-CBC / MODP-1024 (IKEv1)",
      "child": "3DES-CBC / MODP-1024 (PFS: Active)",
      "status": "CRITICAL FAIL",
      "standard": "Banned NIST SP 800-131A"
    },
    {
      "type": "[Link 3] Branch <-> Cloud",
      "ike": "AES-128-CBC / MODP-2048 (IKEv2)",
      "child": "AES-128-CBC / No PFS (Transport)",
      "status": "NON-COMPLIANT",
      "standard": "NIST SP 800-77 \u00a74.2.3"
    },
    {
      "type": "[Network Mesh Aggregate]",
      "ike": "Multi-Peer Overlay (3 Tunnels)",
      "child": "64-bit Extended Sequence Numbers",
      "status": "MONITORED",
      "standard": "RFC 4303 \u00a72.2.1"
    }
  ]
}
```

```
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.